

Internal friction in solids

To cite this article: Clarence Zener 1940 *Proc. Phys. Soc.* **52** 152

View the [article online](#) for updates and enhancements.

You may also like

- [The Internal Friction of Annealed Copper at Low Temperatures](#)
D H Niblett and J Wilks
- [Mechanism of internal friction in Cu–Al–Mn shape memory alloy](#)
K W Yang, E G Jia, J P Shui et al.
- [Evaluation of the Amplitude-Dependent Internal Friction of Materials under Inhomogeneous Strain](#)
Shigeru Asano

INTERNAL FRICTION IN SOLIDS

By CLARENCE ZENER, College of the City of New York

ABSTRACT. The various causes of internal friction in solids are discussed. Internal friction depends in general on thermal currents which occur owing both to the inhomogeneous strains set up by the vibrations in the solid, and to the variations in the elastic constants from point to point in polycrystalline materials. The influence of ferromagnetism also is considered.

§ 1. INTRODUCTION

THE term "internal friction" refers to the capacity of a solid to convert its mechanical energy of vibration into internal energy. This unavoidable degradation of energy manifests itself in many ways. It causes the gradual damping of a vibrating solid even when the solid is so supported as to transfer none of its energy to the surroundings. When such a solid is subjected to a periodic force, this continual change of ordered energy into disordered energy prevents the amplitude of vibration from becoming infinite when the frequency of the applied force approaches a natural frequency of the specimen. Finally, when a solid is coupled electromagnetically to an electrical circuit, it contributes not only to the inductance of the circuit, but also to its resistance.

These various manifestations of internal friction give directly various measures of internal friction. Thus observation of the rate of decay of free vibration gives directly the logarithmic decrement δ . Again, one obtains a second measure from a plot of the amplitude of vibration against frequency of exciting force near a resonance frequency. This is the ratio of the width of this resonance curve at half maximum, $\Delta\nu$, to the frequency at resonance ν . Finally, from the effective resistance R and effective inductance L of the solid coupled electromagnetically to an electrical circuit, one may form a third dimensionless quantity, $2\pi\nu(L/R)$. This ratio is called by electrical engineers the Q of the element. Measures of internal friction have also been used which are obtained from theoretical considerations. If ΔE is the energy dissipated per cycle, and E is the average energy of vibration during the cycle, the ratio $\Delta E/E$ gives a measure of the damping capacity of the solid. This measure is called the specific energy loss. Again, the phenomenon of internal friction may be treated phenomenologically in the theory of elasticity by regarding the elastic modulus as complex. The ratio of the real part of the modulus to the imaginary part is equal to the ratio already denoted by Q . The relation between these various measures is as follows

$$1/Q = \Delta E/2\pi E = \Delta\nu/\sqrt{3}\nu = \delta/\pi. \quad \dots\dots(1)$$

No one of these is used universally. As his measure of internal friction the writer has adopted $1/Q$.

The measured internal friction includes both the energy dissipated by the specimen and also that dissipated to the surroundings. While acoustical losses are

readily eliminated, loss of energy through the supports is especially elusive. Unfortunately many experiments during this century have not profited from the early work of Voigt⁽¹⁾, which showed the extreme care which must be taken to avoid such losses when the internal dissipation is small. A convenient method of avoiding these losses, the one used in this laboratory, is shown in figure 1. The specimen is supported by silk threads at its nodes of vibration. The driving force is obtained by the reaction upon a permanent magnet M of the eddy currents induced by an alternating current in the driving coil. An alternating electromotive force is induced in a detector coil at the other end by the eddy currents arising from the motion of the specimen in the field of a second permanent magnet M' .

The internal friction of solids covers a wide range. Q varies from near unity for highly damped solids to more than 1,000,000 for quartz^(2,3). The value of Q for dry wood⁽⁴⁾ is of the order of 100. Metals cover a wide range, with values of Q

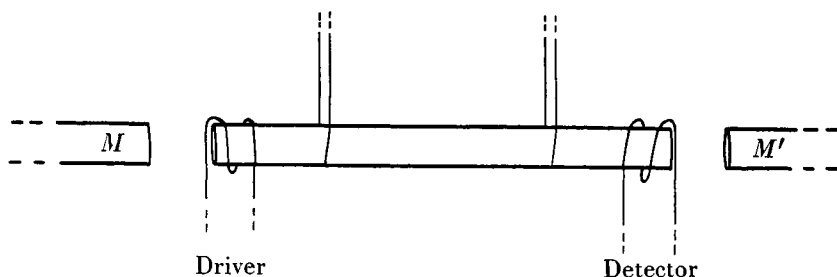


Figure 1. Method of measuring internal friction for strains less than 10^{-5} ; after Randall, Rose and Zener⁽¹⁶⁾.

ranging from less than 100 to more⁽¹⁶⁾ than 200,000. The internal friction of metals is particularly interesting since for any particular metal it varies widely according to the conditions of measurement and to the past history of the specimen.

The internal friction of metals is independent of the amplitude of vibration used in its measurement only if this amplitude is extremely small (strain less than 10^{-5}). A typical example of variation with amplitude is seen in figure 2. In order to avoid this variation the amplitude must be so small that optical methods of detection are too insensitive. Fortunately the modern technique of electrical amplification enables one to measure⁽⁵⁾ the internal friction even when the maximum strain is as low as 10^{-8} . According to Föppl⁽⁶⁾ internal friction of large amplitude is primarily a measure of the capacity of a metal to undergo plastic flow under large stresses. It is hence incapable of yielding information concerning metals not previously disturbed by plastic flow. For this reason this paper is devoted exclusively to internal friction measured with small amplitudes of vibration (maximum strain less than 10^{-5}).

The internal friction of a metal is generally a function of the frequency of measurement^(1,5), while that of dielectrics appears to be independent of this frequency^(4,7). Gemant⁽⁸⁾ has developed a theory of the internal friction of solid dielectrics which interprets this independence of frequency.

The internal friction of a metal increases in a characteristic manner with the temperature of measurement^(9, 10, 11). The type of variation obtained is shown in figure 3. Förster and Köster⁽¹⁰⁾ have attributed this rapid rise with increasing temperature to the onset of slip at high temperatures.

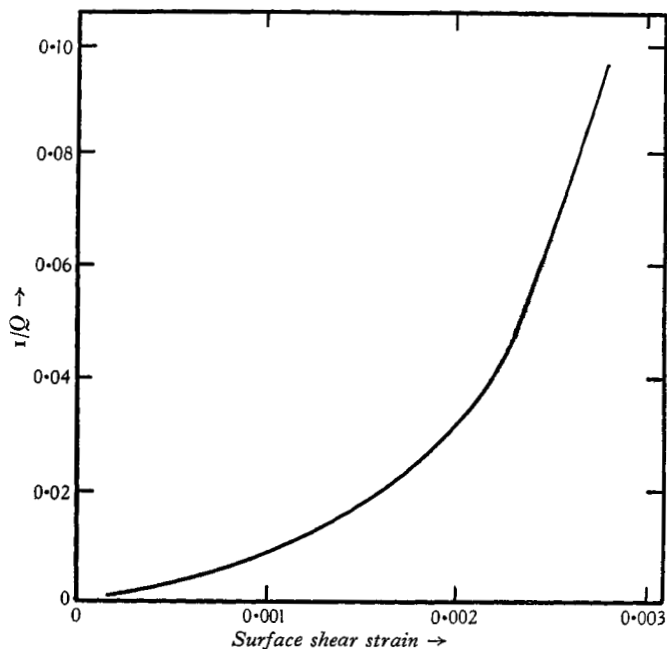


Figure 2. Dependence of internal friction upon amplitude of vibration. Example of torsional vibration of nickel-steel bar, after Foppl⁽⁶⁾.

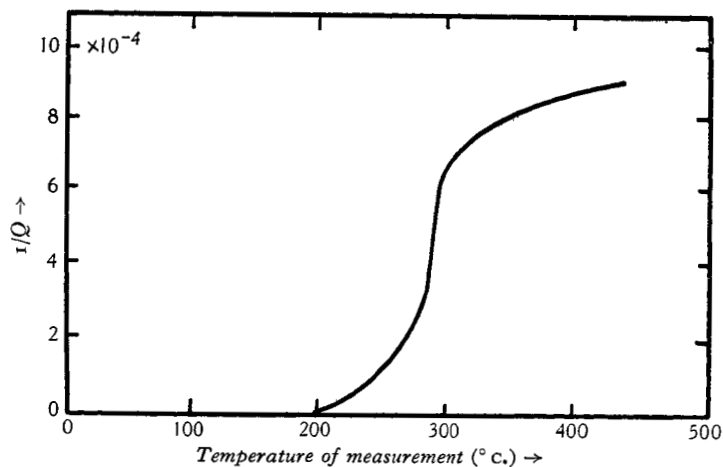


Figure 3. Dependence of internal friction upon temperature of measurement. Example of magnesium, after Forster and Koster⁽¹⁰⁾.

The internal friction of metals is affected more than any other property by the past history of the specimen. A striking example is the variation of the internal friction with the annealing temperature of a previously cold-worked specimen^(5, 10, 11). Figure 4 shows the type of variation found. The damping has been decreased by a factor as large as 30 by annealing to just below the recrystallization temperature. This effect has been interpreted as due to the relief of internal stresses by low-temperature annealing. On the other hand, such annealing has no noticeable effect on the hardness of the specimen, and increases the elastic moduli by at most 5 per cent. The rapid rise upon annealing above the recrystallization temperature has been interpreted as due to an increase in grain size.

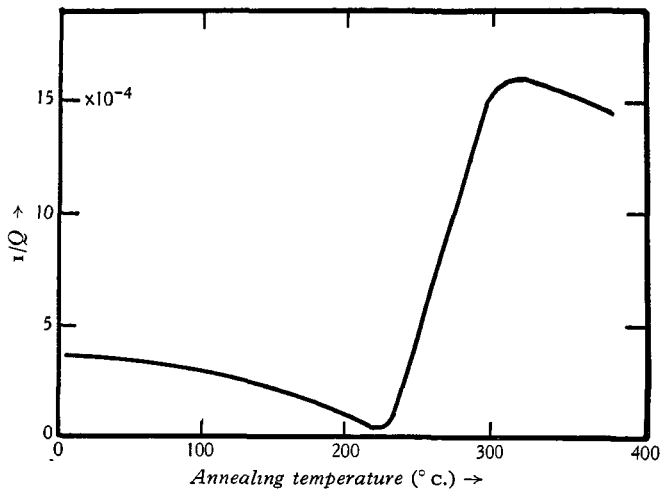


Figure 4. Dependence of internal friction upon annealing temperature. Example of copper, after Wegel and Walther⁽⁵⁾.

§ 2. THERMAL CURRENTS AS A SOURCE OF INTERNAL FRICTION

A. General theory

We have seen that internal friction is an extremely structure-sensitive property. If all the causes of internal friction were thoroughly understood, we could use it as a powerful metallurgical tool. The realization of this potential importance of internal friction has furnished the incentive for the theoretical and experimental work reviewed in this section⁽¹²⁻¹⁶⁾.

From the energetic standpoint, internal friction is simply the transformation of mechanical energy into internal energy. In searching for the causes of this transformation, it is natural to examine first the effects of the direct coupling between the mechanical and the internal coordinates of a solid. This is called the thermoelastic coupling. Its effects are seen in the cooling of a solid during an adiabatic expansion, and conversely, in the expansion of a solid heated under constant pressure. This thermoelastic coupling is operative only when the stresses in the

vibrating solid are inhomogeneous. Periodic changes in stresses during vibration give rise to periodic changes in temperature. If the stresses are not uniform, the temperature will likewise not be uniform. Gradients in temperature give rise to heat currents. But heat currents are necessarily associated with an increase in the entropy of the system. If the entropy of a freely vibrating solid is increasing, its internal energy must be increasing at the expense of its mechanical energy. Hence the thermoelastic coupling endows the solid with internal friction.

An insight into the factors which influence this thermoelastic internal friction may be obtained by considering the flow of heat to and from an idealized region in a specimen undergoing forced vibration. The region will be considered to have such a high thermal conductivity that it is at a uniform temperature. If the region were thermally insulated, its temperature would undergo periodic fluctuations. We shall denote by ΔT the amplitude of this temperature fluctuation. Since there is no flow of heat, the entropy remains unaltered. On the other hand, if it were connected by a perfect thermal conductor to a heat reservoir, vibration would be accompanied by a flow of heat back and forth between the region and the reservoir. We shall denote by $\Delta \mathcal{Q}$ the quantity of heat which thus flows back and forth. Again the entropy remains unaltered, since the flow of heat is between regions with the same temperature. We finally consider the region connected to the heat reservoir by a wall with a finite thermal conductivity. Such a wall introduces a finite time of relaxation τ for the establishment of temperature equilibrium. Thus in the absence of vibration, any temperature difference between our idealized region and the heat reservoir would decay exponentially as $\exp(-t/\tau)$. The now irreversible flow of heat across the temperature gradient of the wall results in a steady increase of entropy. An elementary calculation gives the entropy increase per cycle as

$$\Delta S = \pi T^{-2} \Delta T \Delta \mathcal{Q} \frac{\nu_0 \nu}{\nu_0^2 + \nu^2}. \quad \dots\dots(2)$$

Here ν is the frequency of vibration, and $\nu_0 = 1/(2\pi\tau)$.

This expression for ΔS may be readily interpreted after rewriting it as the product of the two factors $2\Delta \mathcal{Q} \Delta T/T^2$ and $(\pi/2) \nu_0 \nu/(\nu_0^2 + \nu^2)$. The first factor is just the entropy increase which would accompany the flow of heat $\Delta \mathcal{Q}$ from the temperature $T + \Delta T$ to the slightly lower temperature T , and then back again to the still lower temperature $T - \Delta T$. The second factor takes account of the fact that the actual flow of heat is always less than $\Delta \mathcal{Q}$, and that the actual difference of temperature between region and heat reservoir is always less than ΔT . When $\nu \gg \nu_0$, the flow of heat is much less than $\Delta \mathcal{Q}$, and the vibration is nearly adiabatic. When $\nu \ll \nu_0$, the maximum difference in temperature is much less than ΔT , the vibration is nearly isothermal. These two regions are illustrated in figure 5.

The calculation of the entropy increase arising from thermal currents in an actual solid follows closely that for the above idealized system. We now denote by $\Delta T(x, y, z)$ the temperature change in the solid which would result if the vibration proceeded from zero to maximum strain adiabatically, and by $\Delta \mathcal{Q}(x, y, z) dv$ the corresponding heat flow from an element of volume dv if the vibration proceeded

isothermally. The relaxation of any temperature fluctuation in an actual solid may be described in the standard way by an expansion in terms of a suitable infinite set of eigenfunctions. Each such function is associated with a definite time of relaxation. A calculation⁽¹³⁾ similar to that for the idealized system gives as the entropy increase per cycle for the whole specimen an expression similar to equation (2), namely

$$\Delta S = \pi T^{-2} \int \Delta T \Delta \mathcal{Q} dv \sum f_k \frac{\nu_k \nu}{\nu_k^2 + \nu^2}. \quad \text{.....(3)}$$

The frequency ν_k is equal to $1/2\pi\tau_k$, where τ_k is the time of relaxation associated with the k th eigenfunction. The weighting factor f_k satisfies the condition

$$\sum f_k = 1. \quad \text{.....(4)}$$

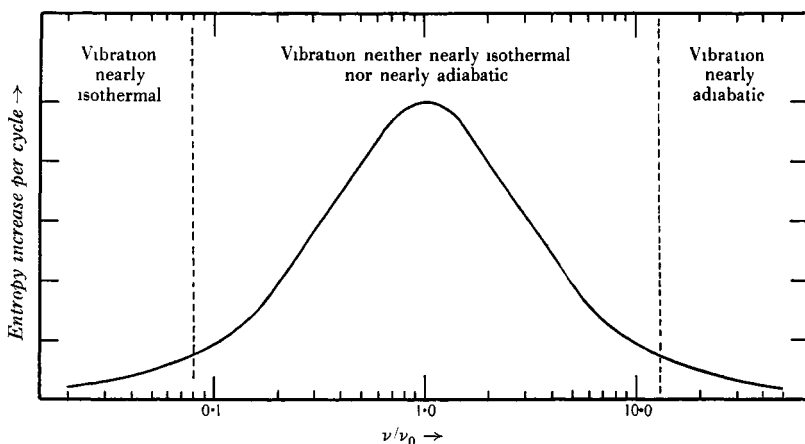


Figure 5. Theoretical dependence of entropy increase per cycle, and of internal friction, with frequency when the thermal diffusion can be described by a single time of relaxation; after Zener⁽¹²⁾.

The gain of internal energy per cycle, ΔE , and hence the loss of mechanical energy per cycle, is $T \Delta S$. The various measures of internal friction may thus be obtained from equation (1). In particular

$$1/Q = \left\{ \frac{T^{-1} \int \Delta \mathcal{Q} \Delta T dv}{2 \times \text{energy of vibration}} \right\} \sum f_k \frac{\nu_k \nu}{\nu_k^2 + \nu^2}. \quad \text{.....(5)}$$

Both $\Delta \mathcal{Q}$ and ΔT are proportional to the amplitude of vibration, while the energy of vibration is proportional to the square of this amplitude. Hence the internal friction resulting from thermal flow is independent of the amplitude of vibration.

B. Macroscopic thermal currents

A pure shear is unaccompanied by any macroscopic thermoelastic effects, as is evidenced by the identity of the adiabatic and isothermal shearing moduli. Hence macroscopic thermal currents are absent in the torsional vibration of cylindrical rods.

On the other hand, pure tensile stresses are associated with comparatively large thermoelastic effects. Pure tensile stresses occur both in longitudinal and in transverse vibrations. In such vibrations the first factor in equation (5) may be reduced by thermodynamics to $(E_S - E_T)/E_S$, where E_S and E_T are the adiabatic and isothermal Young's moduli, respectively. The values of this ratio given in table 1 were calculated from the well-known formula

$$(E_S - E_T)/E_S = TE_T \alpha^2 / \rho C_p,$$

where α is the linear thermal expansion coefficient. Since the E_T of a specimen depends not only upon its composition, but also upon its past treatment, this table must be regarded as only approximate.

In both longitudinal and transverse vibrations, all terms in the summation of equation (5) are negligible compared with the first. In these two cases the internal friction thus has the particularly simple form

$$1/Q = \frac{E_S - E_T}{E_S} \frac{\nu_0 \nu}{\nu_0^2 + \nu^2}. \quad \dots(6)$$

In transverse vibration, the frequency at which the internal friction is a maximum, ν_0 , depends upon the shape and dimensions of the cross-section of the specimen. For a rectangular cross-section

$$\nu_0 = 1.57D/(\text{transverse width})^2; \quad \dots(7)$$

for a circular cross-section

$$\nu_0 = 2.16D/(\text{diameter})^2. \quad \dots(8)$$

In longitudinal vibrations

$$\nu_0 = C^2/2\pi D, \quad \dots(9)$$

where C is the velocity of wave propagation. Values of the thermal diffusion constant D are given in table 1.

Table 1

Metal	D (in cm ² /sec.)	$\frac{E_S - E_T}{E_S}$	Metal	D (in cm ² /sec.)	$\frac{E_S - E_T}{E_S}$
Aluminium	0.88	0.0046	Nickel	0.15	0.0029
Antimony	0.12	0.0018	Palladium	0.26	0.0020
Beryllium	0.53	0.0046	Platinum	0.26	0.0015
Bismuth	0.065	0.0014	Rhodium	0.29	0.00069
Cadmium	0.46	0.010	Silver	1.74	0.0034
Copper	1.2	0.0030	Tantalum	0.22	0.00030
Gold	1.1	0.0017	Tin	0.40	0.0040
Iron	0.20	0.0024	Tungsten	0.61	0.00078
Lead	0.24	0.0025	Zinc	0.41	0.0088
Magnesium	0.60	0.0050	Brass (70-30)	0.38	—

Recent experiments have beautifully demonstrated this damping of transverse vibrations by the flow of heat back and forth across the specimen. Bennowitz and Rötger⁽¹⁷⁾ studied the internal friction of wires, about 2 mm. in diameter, of silver, aluminium, brass, steel and glass. Their results are given in figure 6. These experiments were the first to show a maximum in the internal friction when plotted as a

function of frequency. Although cognizant that these maxima were due to a relaxation phenomenon, they were unaware that this pertained to the temperature fluctuation across the specimens. The position of the maxima as calculated by equation (8), shown as vertical lines in figure 6, is in nearly perfect agreement with the observed maxima⁽¹⁴⁾. On becoming aware of the interpretation of their original experiments, Bennewitz and Rötger then subjected the above theory of damping by thermal currents to a rigorous test by experiments on german silver⁽¹⁸⁾. This metal was chosen to minimize plastic flow. Their own comparison of this experiment with the theoretical curve is given in figure 7. In constructing the theoretical

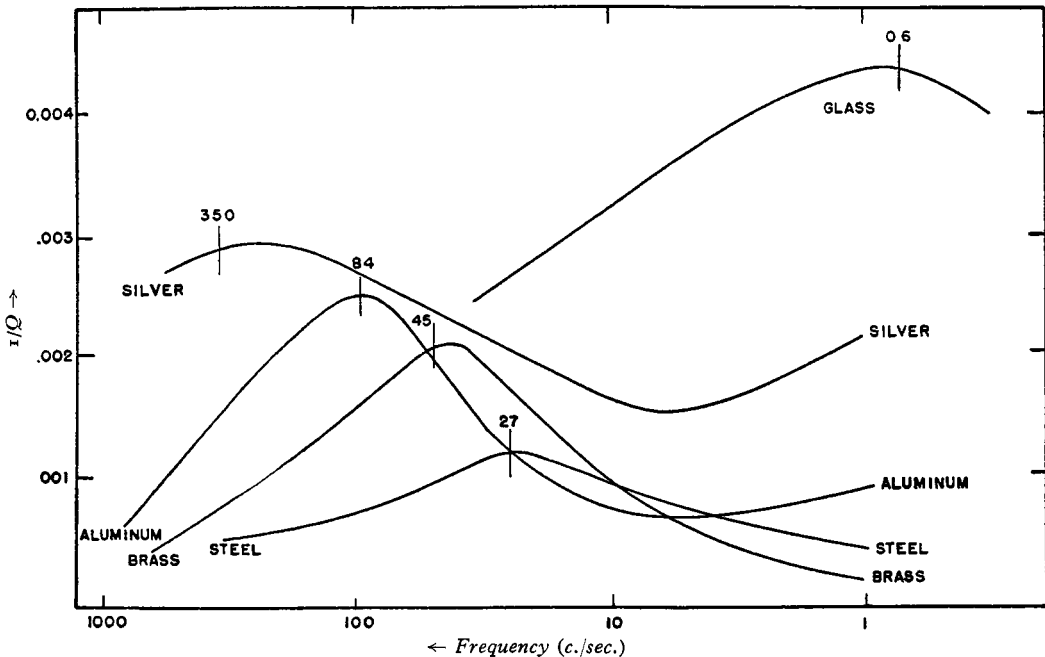


Figure 6. Internal friction of wires vibrating transversely, after Bennewitz and Rotger⁽¹⁷⁾. Positions of maxima calculated by equation (8) are given by vertical lines.

curve, they computed the ratio $(E_s - E_T)/E_s$ for the actual specimens upon which their internal friction measurements were made.

These experiments, as well as similar ones by the author with Otis and Nuckolls⁽¹⁴⁾ with specimens of rectangular cross-section, indicate that over a wide frequency range the damping of transverse vibrations by thermal currents is of a larger order of magnitude than that due to all other causes.

On the other hand, macroscopic thermal currents from regions of compression to regions of dilatation are unimportant in longitudinal vibrations except for exceedingly high frequencies. Thus for copper, equation (9) gives the frequency for maximum internal friction as 1.7×10^{10} c./sec. Longitudinal vibrations at ordinary frequencies are nearly adiabatic with respect to such thermal currents.

Contrary to other examples, the vibrations become less adiabatic as the frequency is increased. This is because the relaxation time for temperature equilibrium is proportional to the square of the wave-length, while the period of vibration is proportional only to the first power of the wave-length.

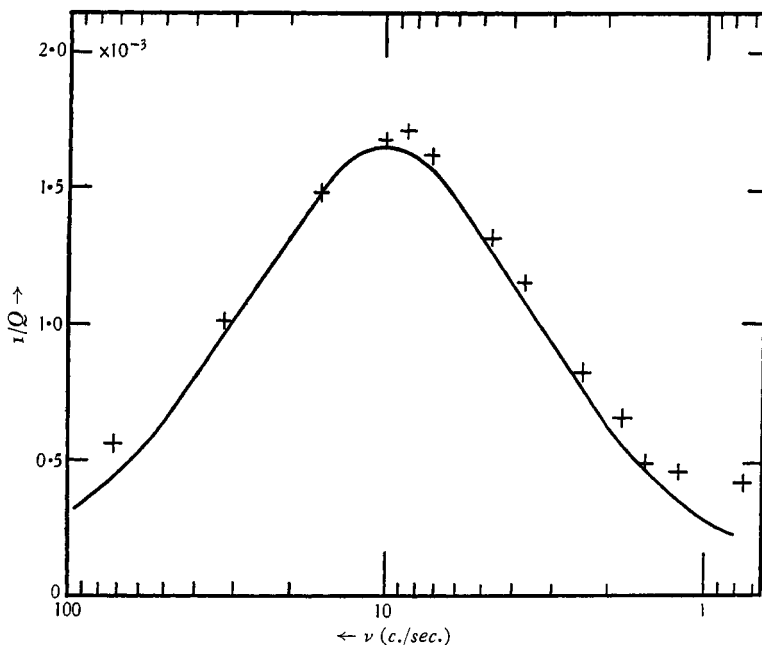


Figure 7. Experimental check of theory of damping of transverse vibrations by thermal currents. Example of german silver, after Bennewitz and Rotger⁽¹⁸⁾. Curve is given by equation (6) with no arbitrary parameters.

C. Microscopic thermal currents

In the previous section we considered the damping by thermal currents arising from stress inhomogeneities inherent in the type of vibration. We now consider the effect of those stress inhomogeneities caused by the deviation of the metal from a perfect single crystal. Of these deviations the polycrystalline nature of a metal is the most striking. The elastic constants of a metal crystal vary markedly with orientation. Examples are given in table 2, column 1, of the ratio of Young's modulus along the [111] direction and along the [100] direction. This elastic anisotropy, together with random orientation, produces microscopic fluctuations in a stress which is macroscopically homogeneous.

The internal friction resulting from the thermal currents between the microscopic stress inhomogeneities is given, at least formally, by equation (5). While an exact calculation of this internal friction is impossible, yet its important features can be obtained from simple considerations⁽¹³⁾. The second factor of Q^{-1} determines the degree of "adiabaticity" of the vibration. This can depend only upon the mean

grain size a of the specimen, its thermal diffusion constant D , and the frequency of measurement ν . The mean grain size is customarily defined by equating a^{-2} to the number of grains visible on a unit surface. Dimensional arguments show that the internal friction can depend upon these three quantities only in the dimensionless product $\nu a^2/D$. When this product is small, the vibration proceeds isothermally, when large it proceeds adiabatically. An estimate of that value of this product for which the internal friction is a maximum may be obtained by replacing the actual temperature fluctuation by one which varies triply periodically with the period $2a$. For such a system $1/\tau = 3D(\pi/a)^2$. Since $1/2\pi\tau$ is the frequency ν_0 at which the internal friction is a maximum, we have the approximate relation

$$\nu_0 a^2/D \simeq 3\pi/2. \quad \text{.....(10)}$$

The integral of the second factor of Q^{-1} with respect to $\log_e \nu$ is independent of the individual coefficients f_k and ν_k . From the value of this integral, namely π , we are able to calculate the area beneath the curve connecting Q^{-1} and

$$\log_{10} (\nu a^2/D).$$

This is

$$\int_{-\infty}^{\infty} Q^{-1} d \log_{10} (\nu a^2/D) = 1.364 \left\{ \frac{T^{-1} \int \Delta T \Delta \mathcal{Q} dv}{2 \times \text{energy of vibration}} \right\}. \quad \text{.....(11)}$$

Rough estimates of this integral have been made⁽¹³⁾, and are given in table 2, column 3.

Table 2

Metal	E_{111}/E_{100} *	$10^3 \times \int Q^{-1} d \log_{10} (\nu a^2/D)$ (damping by intercrystallite thermal currents)	$\left(\frac{d \log G}{d \log T} \right)^{2\dagger}$
Aluminium	1.2	0.025	0.56
Copper	2.8	0.59	0.020
Gold	2.7	0.36	—
Iron (α)	2.1	0.24	0.0081
Lead	3.6	3.0	5.7
Nickel	—	—	0.01
Silver	2.7	0.57	0.060
Tungsten	1.001	0.01	—
Brass (70-30)	3.5	—	—
Brass (β)	8.94	—	—

* All data from E. Schmid and W. Boas, *Kristallplastizität* (Springer, 1935, pp. 200-2), except for lead from reference (36), and for β -brass from reference (34).

† *Handbuch der Experimental Physik*, 5, 251.

The past experimental literature indicates that, apart from possible macroscopic thermal currents, these intercrystalline thermal currents may be the dominant cause of internal friction at room temperature in annealed polycrystalline non-ferromagnetic metals. According to table 2, aluminium should have the least, lead the greatest, internal friction of the common metals. This is in agreement with experiment. When the product $\nu a^2/D$ is small, the internal friction should increase with an increase in grain size. This has been observed by Förster and Köster⁽¹⁰⁾,

and has been interpreted by them as the cause of the rise of internal friction when the annealing temperature is raised above the recrystallization temperature. Finally, when the product $\nu a^2/D$ is large, the internal friction should decrease with an increase in grain size. In particular, single crystals should have a smaller internal friction than polycrystalline specimens. This has been verified by Read⁽¹⁹⁾.

In order to distinguish the internal friction due to intercrystalline thermal currents from that due to other causes, it is necessary to vary both the grain size and the frequency of measurement. This is now being done in this laboratory by Randall, Rose and the writer⁽¹⁶⁾. Measurements on longitudinal vibrations have been made at 12,000 and 36,000 c./sec. on specimens of 70-30 brass ranging in

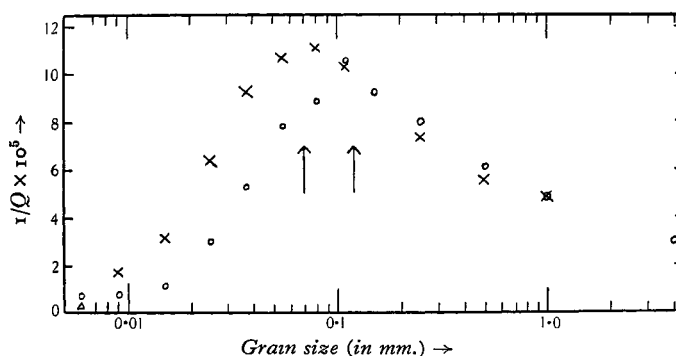


Figure 8. Dependence of internal friction upon grain size. Example of 70-30 brass, after Randall, Rose and Zener⁽¹⁶⁾. Triangle, measurements at 6000 c/sec. Circles, measurements at 12,000 c/sec. Crosses, measurements at 36,000 c/sec. Vertical lines, positions of maxima as calculated by equation (10).

average grain size a from 0.006 mm. to 4 mm. Our present results are shown in figure 8. The experimental curve for each frequency has a maximum at a grain size very near that given by the theoretical equation (10). If the internal friction is actually a function only of the factor $\nu a^2/D$, then shifting the points for 36,000 c./sec. to the right (corresponding to an increase of a by a factor of $\sqrt{3}$) should make these points lie upon the 12,000 c./sec. curve. This they nearly do. The slight discrepancy for grain sizes greater than 0.1 mm. may be due to excessive twinning. From the experimental points an estimate may be made of the integral of Q^{-1} with respect to $\log_{10}(\nu a^2)$. This is of the same order of magnitude as, but 20 per cent smaller than, the estimate given in table 2 for copper.

These experiments furnish convincing evidence of the dominating rôle played by intercrystalline thermal currents in the internal friction of metals. In our extreme isothermal case ($\nu a^2/D = 0.006$), Q^{-1} is less than 0.4×10^{-5} . For the extreme adiabatic case of a single crystal ($\nu a^2/D \sim \infty$) we must take the upper limit of 1.1×10^{-5} for copper obtained by Read⁽¹⁹⁾. In the intermediate range, where the vibration is neither microscopically isothermal nor adiabatic, the internal friction is of the order of 10 times as large, reaching a maximum value of $Q^{-1} = 1.1 \times 10^{-5}$ at $\nu a^2/D = 4.7$. On the other hand, the macroscopic thermal currents accompanying

transverse vibrations give rise to an internal friction larger by still another order of magnitude, reaching a maximum, for brass, of $Q^{-1} = 200 \times 10^{-5}$, as shown in figure 6.

A second way in which actual crystals deviate from perfect single crystals is in the residual stresses introduced by cold working. It has been shown that even in the absence of elastic anisotropy, these residual stresses give rise to thermoelastic damping. The mechanism whereby these thermal currents are generated may be understood by reference to figure 9. Consider two adjacent regions in which the residual shearing stresses are of the opposite sign. If during vibration a uniform shearing stress is superimposed, the total shearing stress of one region will be decreased, that of the other will be increased. The temperature of the first region

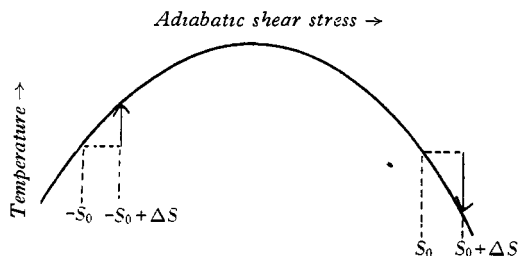


Figure 9. Illustration of thermoelastic damping by residual stresses. Two regions, of shearing stress $-S_0$ and S_0 , are subjected to a uniform shearing stress ΔS . The temperature of the first region increases, that of the second decreases.

will be increased, that of the second decreased. The thermal currents resulting therefrom produce an internal friction whose integral with respect to $\log \nu$ is given approximately by⁽¹⁵⁾

$$\int Q^{-1} d \log_{10} \nu \simeq \frac{\Theta}{T} \left(\frac{d \log G}{d \log T} \right)^2.$$

Here the product of Θ and the specific heat per unit volume is the volume density of energy associated with the residual stresses. It may be as large as $5^\circ \text{C. per c.c.}$ for copper, brass, aluminium, and steel⁽²⁰⁾. G is the shearing modulus. From the values of $(d \log G / d \log T)^2$ given in table 2, we should expect that, in the appropriate frequency range, the thermoelastic internal friction caused by these residual stresses would be comparable to the internal friction due to other causes. The middle of the frequency scale, on a logarithmic scale, will be near D/L^2 , where L is the mean distance between stress inhomogeneities of opposite sign. At present there is no way of estimating this distance.

§ 3 LATTICE DISLOCATIONS AS A SOURCE OF INTERNAL FRICTION

Cold work increases the resistance of a metal to plastic flow, and hence generally reduces the large-amplitude internal friction⁽²¹⁾. On the other hand, it always increases the small-amplitude internal friction, as mentioned in the introduction.

The most striking feature of this cold-work internal friction is its rapid decrease with low-temperature annealing, illustrated in figure 4. The hardening effect of this cold work is not affected by such low-temperature annealing. On the other hand,

the E -modulus is changed considerably by such annealing. At least in steel, the changes of internal friction and of the E -modulus with annealing temperature are closely correlated⁽¹⁰⁾.

The assumption that this correlation is general leads to definite conclusions as to the nature of cold-work internal friction. It at once eliminates the thermoelastic coupling as the primary effect. The calculated difference in the E -modulus for the two extreme cases of microscopically isothermal and microscopically adiabatic vibration is about 100 times smaller than the observed variation of 5 per cent. It rather suggests that the cold working introduces not only lattice distortions, but also lattice imperfections which move under the application of macroscopic stresses, and that it is these imperfections which are removed by low-temperature annealing. Such imperfections are commonly called dislocations⁽²³⁾. The mobility of these dislocations relaxes the rigidity of the metal, and hence lowers the E -modulus. It is to be expected that their motion will be attended by some hysteresis. Their presence thus endows the metal with internal friction. In contrast to thermoelastic internal friction, this internal friction will be observable even at zero frequency, that is, in static stress-strain tests. It may be this introduction of dislocations which causes the static stress-strain curve of zinc single crystals to acquire a hysteresis loop upon being plastically strained⁽²²⁾.

§ 4. FERROMAGNETISM AS A SOURCE OF INTERNAL FRICTION

The anomalously large internal friction of ferromagnetics has been the subject of numerous experiments⁽²⁴⁻³⁰⁾. The source of this internal friction lies in the magnetoelastic coupling of ferromagnetics. In virtue of this coupling, applied stresses change the magnetization, and, conversely, changes in magnetization induce changes in strain. The magnetization of elementary regions is changed by two methods, (*a*) by rotation of the direction of magnetization, and (*b*) by the movement of the boundary between adjacent regions. The latter type of change is accompanied by hysteresis, and hence contributes to the internal friction. All types of change of magnetization are accompanied by electric eddy currents which tend to oppose the change. The irreversibility of these eddy currents also contributes to the internal friction.

A measure of this magnetoelastic coupling is given by the ratio $(E_S - E_0)/E_S$, where E_S and E_0 are Young's moduli at saturation and at zero magnetization, respectively. This ratio has been extensively investigated^(25, 27, 28, 29, 31, 32). It increases with the magnetic softness of the material. For steel it is less than 10^{-4} , for soft iron about 10^{-3} , for permalloy about 10^{-2} . For hard nickel it is 0.007, for soft nickel 0.1, increasing to a maximum of nearly 0.2 at 200° C. The magnetoelastic coupling vanishes at saturation; here small stresses cannot alter the magnetization.

The motion of the boundaries of the elementary magnetic regions may be compared to the motion of dislocations in non-ferromagnetic metals. This motion gives rise to internal friction even at zero frequency, i.e., it produces a hysteresis loop in the static stress-strain curve. This has been beautifully demonstrated by

Becker and Kornetzki⁽²⁶⁾ on a soft iron wire. The damping, as well as the area of the static stress-strain loop, was comparatively large when the iron was unmagnetized. When the magnetoelastic coupling was removed by saturating the iron magnetically, both the internal friction and the area of the hysteresis loop were reduced by a large factor.

The damping by eddy currents may be compared to the damping by thermal currents. The diffusion constant for lines of magnetic induction

$$D_B = 10^8 / (0.4\pi\mu_r\sigma)$$

here plays the same rôle as the thermal diffusion constant D in thermoelastic internal friction. Here μ_r is the reversible permeability, σ is the electrical conductivity in $\text{ohm}^{-1} \text{cm}^{-1}$. These eddy currents may likewise be classified as microscopic and macroscopic. The former are due to the at least partial random orientation of the

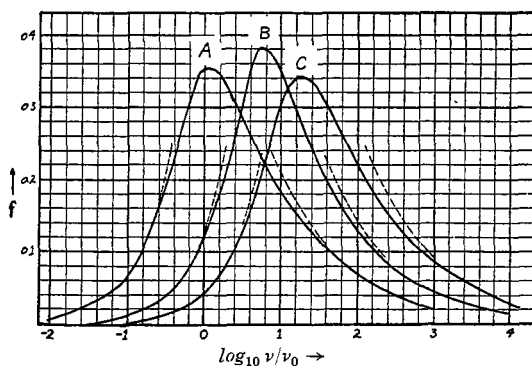


Figure 10. $f(\nu)$ functions associated with macroscopic eddy currents; after Zener³³⁾. $\nu_0 = D_B/2\pi a^2$. Curve A, transverse vibration of reed, a = transverse width of reed. Curves B and C, longitudinal and transverse vibrations of circular rods, a = radius of rod.

magnetization of the elementary regions. Experiments have not yet been performed to separate the effects of these microscopic eddy currents from the effect of magnetic hysteresis.

The macroscopic eddy currents tend to shield the interior of the specimen from changes of magnetic induction⁽³⁵⁾. The internal friction due to these eddy currents is given by⁽³³⁾

$$Q^{-1} = \frac{E_B - E_H}{E_B} f(\nu).$$

Here E_B and E_H are the Young's moduli at constant B and constant H , respectively. The ratio $(E_B - E_H)/E_B$ may be expressed in terms of directly measurable quantities. The plot of the frequency-dependent function $f(\nu)$ is given in figure 10 for various cases. When $\nu \gg \nu_0$ the average magnetic induction remains nearly constant. When $\nu \ll \nu_0$ the average magnetic field strength remains nearly constant. The internal friction due to these macroscopic eddy currents is appreciable only in the intermediate case $\nu \simeq \nu_0$. It has been separated experimentally from that due to magnetic hysteresis only in one particular case⁽²⁹⁾.

REFERENCES

- (1) VOIGT, K. *Ann. Phys., Lpz.*, **47**, 671 (1892).
- (2) VAN DYKE, K. S. *Proc. Inst. Radio Engrs, N. Y.*, **23**, 386 (1935).
- (3) VAN DYKE, K. S. *Phys. Rev.* **53**, 945 (1938).
- (4) KRÜGER, F. and ROHLOFF, E. *Z. Phys.* **110**, 58 (1938).
- (5) WEGEL, R. L. and WALTHER, H. *Physics*, **6**, 141 (1935).
- (6) FÖPPL, O. *J. Iron Steel Inst.* **134**, 393 (1936).
- (7) GEMANT, A. and JACKSON, W. *Phil. Mag.* (7), **23**, 960 (1937).
- (8) GEMANT, A. *Trans. Faraday Soc.* **33**, 1582 (35); **34**, 269 (38); *Physics*, **7**, 311 (1936); *J. Appl. Phys.* **9**, 730 (1938).
- (9) IOKIBE, K. and SALAI, S. *Phil. Mag.* (6), **42**, 307 (1921).
- (10) FÖRSTER, F. and KOSTER, W. *Z. Metallk.* **29**, 116 (1937).
- (11) KOSTER, W. and ROSENTHAL, K. *Z. Metallk.* **30**, 345 (1938).
- (12) ZENER, C. *Phys. Rev.* **52**, 230 (1937).
- (13) ZENER, C. *Phys. Rev.* **53**, 90 (1938).
- (14) ZENER, C., OTIS, W. and NUCKOLLS, R. *Phys. Rev.* **53**, 100 (1938).
- (15) ZENER, C. *Phys. Rev.* **53**, 582 (1938).
- (16) RANDALL, R. H., ROSE, F. and ZENER, C. *Bull. Amer. Phys. Soc.* **14**, No. 2, 8 (1937); *Phys. Rev.* **56** (in the Press).
- (17) BENNEWITZ, K. and RÖTGER, H. *Phys. Z.* **37**, 578 (1936).
- (18) BENNEWITZ, K. and RÖTGER, H. *Z. techn. Phys.* **19**, 521 (1938).
- (19) READ, T. A. *Phys. Rev.* **54**, 389 (1938).
- (20) QUINNEY, H. and TAYLOR, G. I. *Proc. Roy. Soc. A*, **163**, 157 (1937).
- (21) CANFIELD, R. H. *Trans. Amer. Soc. Steel Treat.* **20**, 549 (1932).
- (22) HANSON, A. W. *Phys. Rev.* **45**, 324 (1934).
- (23) HOUWINK, R. *Elasticity, Plasticity, and Structure of Matter*, pp. 114-27. (Camb. Univ. Press, 1937.)
- (24) ZACHARIAS, J. *Phys. Rev.* **44**, 116 (1933).
- (25) V. AUWERS, O. *Ann. Phys., Lpz.*, (5), **17**, 83 (1933).
- (26) BECKER, R. and KORNETZKI, M. *Z. Phys.* **88**, 634 (1934).
- (27) SIEGEL, S. and QUIMBY, S. L. *Phys. Rev.* **49**, 663 (1936).
- (28) COOKE, W. T. *Phys. Rev.* **50**, 1158 (1936).
- (29) BROWN, W. F. *Phys. Rev.* **50**, 1165 (1936).
- (30) WALLER, MARY D. *Proc. Phys. Soc.* **50**, 144 (1938).
- (31) ENGLER, O. *Ann. Phys., Lpz.*, (5), **31**, 145 (1938).
- (32) NAKAMURA, K. *Z. Phys.* **94**, 707 (1937).
- (33) ZENER, C. *Phys. Rev.* **53**, 1010 (1938).
- (34) WEBB, W. *Phys. Rev.* **55**, 297 (1939).
- (35) KERSTER, M. *Z. techn. Phys.* **15**, 463 (1934).
- (36) GOENS, E. *Wiss. Abh. Phys.-Techn. Reichsanstalt*, **20**, 63 (1936).